

A candidate 67.302666% lower bound for simple critical-line zeros of the Riemann zeta function

AI-assisted derivative research artifact. Curated and released by Uwe Schwarz. Mathematical provenance and contribution boundaries are stated in the Reproducibility section.

Abstract

Let $N(T, 2T)$ count the zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. This derivative artifact updates the finite certificate in Ainta’s AI-assisted draft [3]. It retains Ainta’s stability refinement, seven-point use of the Montgomery–Taylor overlap kernel inherited from Claude’s analysis, block aggregation, shifted pinching, and verifier lineage [1]. We add an exact finite-dimensional spectral envelope and a stronger rational interval certificate, obtaining the candidate unconditional bound

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6730266625438475496579 \dots$$

The finite certificate proves an exact seven-point inequality at the rational target $382623/10^8$. Its standalone Arb replay is complete. A substantial Lean formalization has been kernel-checked, but the dependency-closed build of all generated terminal catalogs and the final public theorem is not complete; this paper is therefore released as an AI-assisted, unreviewed research draft, not as a completed formal-verification claim or a numerical-record claim.

1 Statement and setup

We use the notation and normalisations of [1, §§2,4,7.1]. Put

$$\ell := \log \frac{T}{2\pi}, \quad I := (T, 2T], \quad D_0 := T^{1/2}, \quad I' := (T - D_0, 2T + D_0].$$

For the optimised test family of Theorem D, $L = \ell$ and

$$\phi(u) = \cos\left(\frac{\sqrt{2}u}{\ell}\right)^{1/2} \varrho(L/2 - |u|),$$

where the fixed-width ramp ϱ is defined in [1, §2.2]. Let

$$a := \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du, \quad \hat{G} := \frac{G}{aL^2}, \quad \hat{A} := \frac{A}{aL^2}, \quad d := \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

As in [1, §4.1], divide the zeros in I' into the simple critical-line zeros $\mathcal{S}_1$, the distinct multiple critical-line zeros $\mathcal{S}_2$, and the unordered off-line pairs $\mathcal{P}$. Write

$$s_1 := \#\mathcal{S}_1, \quad s_2 := \#\mathcal{S}_2, \quad p := \#\mathcal{P}.$$

Then

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{1}$$

For $\rho \in \mathcal{S}_1$, define

$$u_\rho := (\hat{\phi}(\gamma_\rho - \tau_k))_{0 \leq k < d} \in \mathbb{R}^d, \quad v_\rho := \frac{u_\rho}{\sqrt{a}L}.$$

The simple-zero part of $\hat{A}$ is

$$P_1 = VV^\top = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^\top,$$

where the columns of V are the v_ρ . By Lemma 2.2 of [1], $\|v_\rho\| \leq 1$. For

$$Q' := \hat{A} - P_1,$$

the decomposition in the proof of Proposition 4.4 of [1] gives

$$n_+(Q') \leq s_2 + p. \tag{2}$$

Set

$$H_{\text{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457 \dots$$

Theorem D of [1] proves the lower bound H_{MT} . The updated finite input gives the following revision of Ainta's bound.

Theorem 1.1. *One has*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{864900000000H_{\text{MT}} - 1723600000}{867988393769 - 20000\sqrt{100022498758}} = 0.6730266625438475496579 \dots$$

Theorem 1.1 uses Proposition 4.1 as an external computer-assisted input. That proposition is supported here by the standalone Arb certificate described in Section 4. The partial Lean consumer does not yet provide a dependency-closed proof of the theorem.

2 A stability refinement of the rank–trace inequality

Define the convex function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases} \quad (3)$$

For a positive semidefinite matrix M , let

$$\mathcal{D}(M) := \operatorname{tr} \Psi(M).$$

Lemma 2.1 (Stability-enhanced rank–trace inequality). *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$, and let Q be Hermitian with $n_+(Q) \leq b$. Then*

$$\|P + Q\|_{\text{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \quad (4)$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_{\pm} \succeq 0$, $Q_+Q_- = 0$, and $\operatorname{rank} Q_+ \leq b$. Since $P, Q_+ \succeq 0$,

$$\|P + Q\|_{\text{F}}^2 \geq \|P - Q_-\|_{\text{F}}^2 + \|Q_+\|_{\text{F}}^2.$$

If $q_j > 0$ are the nonzero eigenvalues of Q_+ , then $q_j^2 \geq 4q_j - 4$, so

$$\|Q_+\|_{\text{F}}^2 \geq 4 \operatorname{tr} Q_+ - 4b. \quad (5)$$

Let $p_1 \geq \dots \geq p_r \geq 0$ be the eigenvalues of M (the nonzero eigenvalues of P , padded by zeros), and let $n_1 \geq n_2 \geq \dots \geq 0$ be the eigenvalues of Q_- . Von Neumann's trace inequality gives $\operatorname{tr}(PQ_-) \leq \sum_{i \leq r} p_i n_i$, hence

$$\|P - Q_-\|_{\text{F}}^2 + 4 \operatorname{tr} Q_- \geq \sum_{i \leq r} ((p_i - n_i)^2 + 4n_i).$$

For $p \geq 0$,

$$\min_{n \geq 0} ((p - n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_{\text{F}}^2 \geq 2 \operatorname{tr} P - r + \mathcal{D}(M) - 4 \operatorname{tr} Q_-. \quad (6)$$

Combining (5) and (6) gives

$$\|P + Q\|_{\text{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 2 \operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Since $\operatorname{tr} P$ is the sum of the squared column norms of V , $\operatorname{tr} P \leq r$. □

Corollary 2.2. *With $M := V^{\top}V$,*

$$s_1 \geq 4 \operatorname{tr} \hat{A} - \|\hat{A}\|_{\text{F}}^2 - 2N(I') + \mathcal{D}(M). \quad (7)$$

After removing the tails as in Proposition 4.2 of [1],

$$N_0^s(T, 2T) \geq H_{\text{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)), \quad (8)$$

where M° is the Gram matrix of the retained central simple zeros.

Proof. Apply Lemma 2.1 to $P_1 + Q' = \hat{A}$. Equations (1) and (2) give

$$\|\hat{A}\|_{\text{F}}^2 \geq 4 \operatorname{tr} \hat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \geq 4 \operatorname{tr} \hat{A} - s_1 - 2N(I') + \mathcal{D}(M),$$

which is (7).

Use the same comparison $\hat{A} = \hat{G} - \hat{E}$ and tail estimates as in Propositions 4.2 and 4.4 of [1]. Pinching the full Gram matrix into the retained central block and its complement gives $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$ because $X \mapsto \operatorname{tr} \Psi(X)$ is convex and unitarily invariant. Moreover $N(I') = N(T, 2T) + o(N(T, 2T))$, and Theorem D gives

$$\operatorname{tr} \hat{G} = N(T, 2T)(1 + o(1)), \quad \|\hat{G}\|_{\text{F}}^2 = \left(\frac{1}{c_1^*} + o(1) \right) N(T, 2T),$$

where $2 - 1/c_1^* = H_{\text{MT}}$. Substitution in (7) yields (8). □

3 The Montgomery–Taylor overlap kernel

Put $h := 2\pi/L$ and define the normalised ordinate

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Lemma 3.1. *Fix $R_0 > 0$. Delete the simple zeros in strips of normalised width L^2 at the two ends of the grid. The number deleted is $o(N(T, 2T))$, and uniformly for retained simple zeros with $|x_\rho - x_{\rho'}| \leq R_0$,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \quad (9)$$

where

$$k(x) := \frac{K(x)}{K(0)}, \quad (10)$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \end{aligned} \quad (11)$$

and $K(0) = \sqrt{2} \sin(1/\sqrt{2})$. The singularities in (11) are removable.

Proof. For the full Gabor grid, Lemma 2.2 of [1] gives

$$\frac{1}{aL^2} \sum_{j \in \mathbb{Z}} \hat{\phi}(\gamma - \tau_j) \hat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \quad \Phi = \widehat{\phi^2}.$$

For points at normalised distance at least L^2 from the grid ends, the terms with $j \notin [0, d]$ contribute $O(L^{-2})$, uniformly when $|x_\rho - x_{\rho'}|$ is bounded, by the r^{-2} decay used in [1, §5.3].

For fixed x , substitution $u = Lt$ gives the normalised full-grid overlap

$$\frac{\Phi(hx)}{aL} = \frac{\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi i x t} dt}{\int_{-1/2}^{1/2} \phi(Lt)^2 dt} \rightarrow \frac{K(x)}{K(0)}. \quad (12)$$

Indeed, $\phi(Lt)^2 \rightarrow \cos(\sqrt{2}t) \mathbf{1}_{[-1/2, 1/2]}(t)$ in L^1 ; the fixed-width taper has rescaled length $O(L^{-1})$. The convergence in (12) is uniform for x in compact sets. Together with the tail estimate, this proves (9). Finally, the deleted strips have ordinate length $O(L)$ and contain $O(L^2) = o(N(T, 2T))$ zeros, since $N(t+1) - N(t) \ll \log(t+3)$. $\square$

Set

$$w(x) := k(x)^2.$$

4 A seven-point local inequality

For nonnegative gaps $g_1, \dots, g_6$, define

$$\mathcal{F}_6(g_1, \dots, g_6) := \frac{1}{3000} \sum_{i=1}^6 g_i + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}). \quad (13)$$

The second term contains all 21 pairwise separations among seven ordered points.

Proposition 4.1 (Certified local inequality). *For every $g_1, \dots, g_6 \geq 0$,*

$$\mathcal{F}_6(g_1, \dots, g_6) \geq \frac{382623}{100000000}. \quad (14)$$

Verification. The accompanying verifier derives from the public implementation in [3] and uses Arb interval arithmetic through `python-flint`. It evaluates the entire-sinc expression

$$K(x) = \frac{1}{2} \left(\frac{\sin(\pi x - 1/\sqrt{2})}{\pi x - 1/\sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{\pi x + 1/\sqrt{2}} \right)$$

at 128-bit precision and constructs rigorous lower bounds for w on a grid of mesh $1/4000$. If the linear pressure alone reaches the target, the box is immediately closed. On the remaining compact region, the one-variable term

$$U(g) := \frac{g}{3000} + \frac{1}{3}w(g)$$

reduces each coordinate to certified unions of grid cells. The resulting boxes are proved by interval lower bounds, certified tangent and convexity bounds, a bounded subcell tangent tree for the hardest cases, or bisection of the widest coordinate. The program fails if a terminal box remains unresolved.

The committed certificate records `verified=true`, grid 4000, precision 128 bits, 980,069 visited nodes, 490,399 pruned nodes, 489,670 splits, and maximum depth 65. The terminal breakdown is 285,258 interval, 3,166 pressure, and 201,975 tangent terminals; 1,773 of the tangent terminals use the bounded subcell fallback. The report also records the hashes of the tables and certificate data used in the search. $\square$

Lemma 4.2. *Let $y_1 < \dots < y_m$ with $m \geq 7$, and put*

$$E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i).$$

Then

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{382623}{100000000}(m - 6). \quad (15)$$

Proof. Write $g_i = y_{i+1} - y_i$ and sum (14) over the $m - 6$ consecutive seven-point windows. A pair spanning r gaps occurs at most $7 - r$ times and has coefficient $2/(7 - r)$ each time, so its total contribution is at most $2w(y_j - y_i)$. Each single gap occurs at most six times, giving at most

$$\frac{6}{3000} \sum_{i=1}^{m-1} g_i = \frac{1}{500}(y_m - y_1).$$

$\square$

Lemma 4.3 (Exact spectral envelope). *Let G be an $m \times m$ correlation matrix, and put*

$$E := \text{tr}(G - I)^2 = 2 \sum_{i < j} |G_{ij}|^2, \quad D := \text{tr} \Psi(G).$$

For $0 \leq E \leq m(m-1)$, define

$$g_m(E) := \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & m/(m-1) \leq E \leq m(m-1). \end{cases} \quad (16)$$

Then $D \geq g_m(E)$. Moreover, g_m is increasing and 1-Lipschitz.

Proof. Write the eigenvalues of G as $1 + y_i$. Then $y_i \geq -1$, $\sum_i y_i = 0$, and $E = \sum_i y_i^2$. If every $y_i \leq 1$, then $D = E$. Otherwise let $H = \{i : y_i > 1\}$, put $h = |H|$, write $y_i = 1 + u_i$ on H , and set

$$q_+ = \sum_{i \in H} u_i, \quad r^2 = \sum_{i \in H} u_i^2, \quad R = \sum_{i \notin H} y_i^2.$$

The complementary deviations sum to $-(h + q_+)$, so Cauchy's inequality and $q_+ \geq r$ give

$$\begin{aligned} E &\geq h + 2r + r^2 + \frac{(h+r)^2}{m-h} \\ &= \frac{m(1+r)^2}{m-1} + \frac{(h-1)(m+r)^2}{(m-h)(m-1)}. \end{aligned}$$

Since $E - D = r^2$, this implies

$$D \geq E/m + 2\sqrt{(m-1)E/m} - 1.$$

It also shows that the second case requires $E > m/(m-1)$. Equality is realized by the correlation matrix

$$\left(1 - \frac{a}{m-1}\right) I + \frac{a}{m-1} J, \quad a = \sqrt{(m-1)E/m}.$$

On the second branch,

$$g'_m(E) = \frac{1}{m} + \sqrt{\frac{m-1}{mE}} \leq 1,$$

which also proves monotonicity and the Lipschitz assertion. $\square$

Take

$$m := 279, \quad A := \frac{382623}{100000000}(m-6) = \frac{104456079}{100000000}, \quad (17)$$

and put

$$C := g_{279}(A) = \frac{11606231}{3100000000} + 2\sqrt{\frac{1613266109}{1550000000}} - 1. \quad (18)$$

If $e + x \geq A$, monotonicity and the 1-Lipschitz property imply

$$g_m(e) + x \geq g_m(A) = C. \quad (19)$$

Indeed, this is immediate for $e \geq A$, while for $e < A$ one has $x \geq A - e \geq g_m(A) - g_m(e)$.

Let B be a block of 279 consecutive retained simple zeros, and write G_B for its Gram matrix. If its span term already exceeds A , the following estimate is immediate because $C \leq A$. Otherwise the

span is bounded, so Lemma 3.1 applies uniformly to every pair. The diagonal entries are $1 + o(1)$ uniformly. To make the normalization explicit, put

$$\delta_T := \max_i |(G_B)_{ii} - 1| = o(1), \quad D_B := \text{diag}((G_B)_{ii}^{-1/2}), \quad R_B := D_B G_B D_B.$$

The matrix R_B is a correlation matrix. Since m is fixed and the Gram entries satisfy $|(G_B)_{ij}| \leq \sqrt{(G_B)_{ii}(G_B)_{jj}}$, one has $\|G_B\|_F = O_m(1)$ and

$$\|R_B - G_B\|_F \leq \|D_B - I\|_{\text{op}} \|G_B\|_F \|D_B\|_{\text{op}} + \|G_B\|_F \|D_B - I\|_{\text{op}} = O_m(\delta_T) = o(1).$$

The Hoffman–Wielandt inequality and the global 2-Lipschitz property of Ψ now give

$$|\mathcal{D}(R_B) - \mathcal{D}(G_B)| \leq 2\sqrt{m} \|R_B - G_B\|_F = o(1),$$

while

$$\left| \sum_{i \neq j} |(R_B)_{ij}|^2 - \sum_{i \neq j} |(G_B)_{ij}|^2 \right| \leq (\|R_B\|_F + \|G_B\|_F) \|R_B - G_B\|_F = o(1).$$

Thus the normalization changes the defect and off-diagonal energy by $o(1)$. It follows from Lemmas 4.2 and 4.3, together with (19), that uniformly in B ,

$$\mathcal{D}(G_B) + \frac{1}{500} \text{span}(B) \geq C - o(1). \quad (20)$$

5 Shifted block pinching

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalised ordinates of the retained central simple zeros. Then

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)). \quad (21)$$

For each of the $m = 279$ offsets, partition the ordered points into full consecutive blocks of size m , leaving at most $2(m - 1)$ endpoint points. Pinching M° to the full principal blocks and the remaining block gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \quad (22)$$

This follows because pinching is an average of unitary conjugations and $X \mapsto \text{tr } \Psi(X)$ is convex and unitarily invariant.

For each offset, sum (20) over the full blocks and use (22); then average over the m offsets. The average number of full blocks is $S^\circ/m + O(1)$. Each gap $x_{j+1} - x_j$ belongs to a block span for at most $m - 1$ offsets, and

$$x_{S^\circ} - x_1 \leq \frac{T}{h} = \frac{LT}{2\pi} = d + O(1) = N(T, 2T) + o(N(T, 2T))$$

by the Riemann–von Mangoldt formula. The blockwise $o(1)$ term in (20) is uniform, so its total is $o(N(T, 2T))$. Therefore

$$\mathcal{D}(M^\circ) \geq \frac{C}{m} N_0^s(T, 2T) - \frac{m-1}{500m} N(T, 2T) - o(N(T, 2T)). \quad (23)$$

With $m = 279$,

$$\mathcal{D}(M^\circ) \geq \frac{C}{279} N_0^s(T, 2T) - \frac{139}{69750} N(T, 2T) - o(N(T, 2T)). \quad (24)$$

Substitute (24) into (8). This gives

$$\begin{aligned} N_0^s(T, 2T) &\geq H_{\text{MT}} N(T, 2T) + \frac{C}{279} N_0^s(T, 2T) \\ &\quad - \frac{139}{69750} N(T, 2T) - o(N(T, 2T)). \end{aligned}$$

Hence

$$\left(1 - \frac{C}{279}\right) N_0^s(T, 2T) \geq \left(H_{\text{MT}} - \frac{139}{69750}\right) N(T, 2T) - o(N(T, 2T)).$$

Divide by $N(T, 2T)$ and let $T \rightarrow \infty$ to obtain

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{H_{\text{MT}} - 139/69750}{1 - C/279} = 0.6730266625438475496579 \dots$$

Clearing rational factors gives the exact expression stated in Theorem 1.1. Directed 256-bit Arb arithmetic separates this value between decimals ending in $\dots 0978$ and $\dots 0979$. A directed interval-certified scan through $7 \leq m \leq 1000$ gives a unique maximum at $m = 279$, with $m = 278$ second. For completeness, the infinite tail is controlled as follows. Put $q = 382623/10^8$ and, for real $x \geq 1000$, set

$$s(x) := \sqrt{q \left(x - 7 + \frac{6}{x}\right)}.$$

For the continuous version of the second spectral branch,

$$\frac{C_x}{x} = \frac{q(x-6)}{x^2} + \frac{2s(x)-1}{x}.$$

The first term decreases for $x > 12$. The derivative of the second term has the sign of

$$s(x) - q\left(x - 14 + \frac{18}{x}\right).$$

At $x = 1000$, exact rational arithmetic gives

$$q(1000-14)^2 - 1000 = \frac{67996137527}{25000000} > 0, \quad 2q(1000-14) - 1 = \frac{163633139}{25000000} > 0.$$

Consequently $q(x-14)^2 - x$ is positive and increasing for $x \geq 1000$, and

$$s(x) < \sqrt{qx} < q(x-14) < q\left(x - 14 + \frac{18}{x}\right).$$

Thus C_x/x decreases. The numerator $H_{\text{MT}} - 1/500 + 1/(500x)$ also decreases, while the positive denominator $1 - C_x/x$ increases. Hence B_x decreases for $x \geq 1000$, making $m = 279$ the unique optimum over every integer $m \geq 7$. This proves Theorem 1.1.

Reproducibility

Proposition 4.1 is the finite computer-assisted input. The accompanying artifact contains its source, tests, trust-base description, and certificate. The verifier reconstructs every transcendental interval enclosure from the stated formulas; a successful replay reports the exact target $382623/100000000$ and `verified=true`.

The release snapshot uses macOS 26.6.1 on arm64, Python 3.11.15, and `python-flint` 0.9.0, which is the Python binding used to generate the Arb enclosures. The complete release file identity is recorded in `RELEASE_MANIFEST.sha256`; the fresh command and output are recorded in `REPRODUCIBILITY.md`.

The artifact also contains source and partially completed kernel evidence for a Lean certificate consumer. At the snapshot date, the generic mean-value-theorem, segment, replay, and scaled-checker semantics and the complete bounded-subcell adaptive tangent catalog have been kernel-built. The complete 5,793-module mean-value interval terminal catalog has also been kernel-built, transferred, and bound by a common macOS/Linux artifact digest. The full tangent-and-convexity terminal catalog and final dependency-closed public build are deferred. Accordingly, this draft does not claim a completed transitive Lean verification. The machine-readable status matrix and the final gates are recorded in `VERIFICATION_STATUS.md`.

This work is AI-assisted and unreviewed. It is a derivative of the earlier public draft [3]; its stability refinement, seven-point use of the Montgomery–Taylor overlap kernel, block aggregation, shifted pinching, and verifier lineage are retained here. The analytic foundation and limiting overlap kernel come from Claude’s paper and Lean artifact [1, 2]. The new contribution in this revision is the exact finite-dimensional spectral envelope, the stronger exact finite certificate and its reproducibility package, and the partially completed Lean certificate-consumer effort. The clean-room reproduction [4] is an independent comparison only; no proof text or implementation is imported from it. The numerical statement should be described as a candidate improvement within this certificate lineage, not as an established record, until independent review has taken place.

References

- [1] Claude, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, preprint, August 10, 2026, available online.
- [2] Anthropic, *Lean artifact for More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, version 1.0, commit `3635e74826a4c1fcede7d1cd2b6fa75e43a00510`, August 10, 2026, GitHub repository.
- [3] Ainta, *More than 67.30085% of the zeros of the Riemann zeta function are simple and lie on the critical line*, AI-assisted research draft and verifier artifact, commit `040c5e899e658aed7b56a2a87f501798fe10761d`, August 11, 2026, GitHub repository at the cited revision.
- [4] L. Rademacher, *AI refines an AI-derived zeta bound*, clean-room research draft and reproduction artifact, commit `bd4a7d36988b23220034527881f4457f2f689e86`, August 11, 2026, GitHub repository at the cited revision.
- [5] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292.
- [6] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in: Analytic Number Theory (St. Louis, 1972), Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., 1973, 181–193.
- [7] H. L. Montgomery, *Distribution of the zeros of the Riemann zeta function*, in: Proceedings of the International Congress of Mathematicians (Vancouver, 1974), Vol. 1, Canad. Math. Congress, 1975, 379–381.